

TB Ring Modulation

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Overview

Adds metallic resonance and rhythmic trembling with an internal carrier tone.

What this plug-in solves

Ring modulation is powerful but often slowed by external carrier routing. TB Ring Modulation supplies its own carrier, covering tremolo, pulses, robotic speech, bell-like side tones, and extreme metallic textures with four primary sound controls.

What you can expect

- Slow tremolo and pulsing amplitude movement
- Metallic sum-and-difference sidebands
- Robotic voices and tuned percussion
- Simple automation without external routing

How it works

TB Ring Modulation combines the input with an internal oscillator to create new frequencies above and below the original tone. Slow oscillator movement can feel like tremolo, while faster settings produce bell-like, robotic, or metallic side tones.

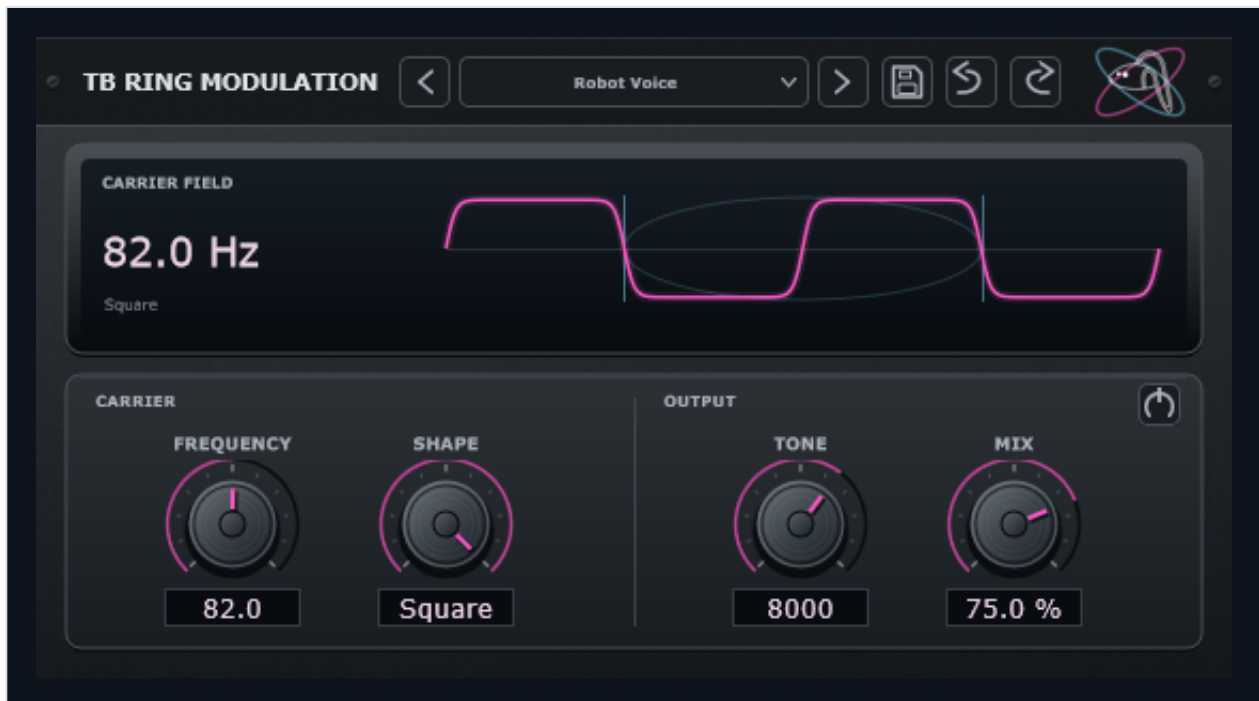
Frequency determines the speed or pitch of the transformation, Shape changes the oscillator from smooth toward harder-edged motion, and Tone controls the brightness of the processed path. Use Mix to preserve the source when tuning is not meant to be exact. Small frequency moves can create large changes in character.

Quick start

- 1 Set Mix fully wet temporarily so the ring-modulated character is easy to hear.
- 2 Sweep Frequency from sub-audio into the audible range.
- 3 Choose Shape for sideband density.
- 4 Use Tone to control harshness.
- 5 Reduce Mix to restore the original note.

- 6 Automate Frequency slowly for transitions or rhythmically for designed motion.

Interface and key sections



This is a direct capture of the current plug-in interface. Use the visible labels to locate the areas and controls explained below.

Frequency

Below the audible range, Frequency behaves like tremolo. As it enters the audio range, new sum-and-difference sidebands become the main effect.

Shape

Moves the carrier waveform from Sine toward progressively harder shapes in the deployed control mapping. Smoother shapes produce cleaner sidebands; harder shapes add more harmonic families.

Tone

Sets the upper tonal focus, controlling edge and brightness after modulation.

Mix

Blends the ring-modulated signal with the original. Partial Mix preserves pitch identity; full wet exposes the sideband structure.

Programs and smoothing

Programs cover Subtle Tremolo, Bell Alloy, Robot Voice, Radio Sidebands, and Digital Insect. Parameter and bypass smoothing reduce clicks during automation.

Controllable properties

The guide uses the names shown on the plug-in panel. Each explanation describes the audible result, a useful way to approach the control, and an important interaction to listen for.

Control	What it does and when to use it
Bypass	Turns the complete effect off or on for a direct before-and-after comparison. Compare with bypass at a similar loudness. If the effect creates a tail, let that tail settle before judging the difference.
Frequency	Chooses the main tonal area or carrier pitch used by this control. Sweep broadly to find the useful tonal area, then make the focus narrower and the change smaller while listening in the full mix.
Mix	Blends the original sound with the processed sound. Turn the processed sound up temporarily to learn its character, then return to the mix and keep only the amount that supports the source.
Program	Loads a factory starting point. Adjust the controls afterward to fit the current sound. Use a preset as a direction, not a finished answer. Change one section at a time so it is clear which adjustment improves the source.
Shape	Moves the internal carrier from a smooth sine wave toward a sharper square wave. Compare the available characters on the same short passage. Choose by the musical result rather than by the name of the mode.
Tone	Darkens or brightens the metallic side frequencies created by the effect. Sweep broadly to find the useful tonal area, then make the focus narrower and the change smaller while listening in the full mix.

Practical uses

Subtle tremolo

- Set Frequency below about 20 Hz.
- Use a smooth Shape.
- Set Mix to taste and leave Tone open.

Expected result: The level pulses without generating a strongly metallic spectrum.

Robot voice

- Set Frequency in the low-to-mid audio range.
- Use a harder Shape.
- Lower Tone if consonants become sharp.
- Keep Mix high.

Expected result: Speech retains rhythm while pitch becomes mechanical and sideband-rich.

Common pitfalls

Ring modulation is inharmonic unless carrier and source relationships align. Return the strongest control toward neutral, compare with bypass at a similar loudness, and add the processing back one section at a time.

High frequencies and hard shapes can be bright. Return the strongest control toward neutral, compare with bypass at a similar loudness, and add the processing back one section at a time.

The internal carrier does not track source pitch. Feed the plug-in one clear note at a time, confirm the musical key or target, and reduce correction or variation if the result becomes unstable.